

Engin Öksüz

AUTONOMY ENGINEER – PERCEPTION & LOCALIZATION

Sancaktepe, İstanbul, Türkiye

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Summary

- Autonomy / robotics software engineer with **6+ years** developing and deploying **perception, visual localization, sensor fusion, and AI-based autonomy** for UAVs and UGVs. Bring research-grade vision and localization methods into real-time robotic systems using **C++**, Python, ROS 2, CUDA, TensorRT, and NVIDIA Jetson.
- On the ground, GNSS-denied localization at **sub-20 cm** and real-time **SAM2** mask tracking at **25 FPS**; in the air, map-aided UAV localization with **14–16 m** bias-free track MAE on **24–25 km** field sorties.

Technical Skills

- Focus:** Visual localization, state estimation, multi-modal sensor fusion, 3D perception, semantic mapping, traversability analysis
- Programming:** C++, Python, Bash
- Robotics:** ROS 1/2, Autoware, MAVROS, Gazebo, RViz, Docker, Git, Ubuntu Linux
- Perception / ML:** PyTorch, OpenCV, PCL, Open3D, SuperGradients, DINOv2, SAM2, RF-DETR, YOLO, ONNX Runtime, TensorRT, CUDA
- Autonomy:** GNSS/GPS-based navigation, GNSS-denied navigation, SLAM, path planning, visual-inertial geo-localization, collision avoidance
- Sensors / hardware:** LiDAR (Velodyne, Robosense), cameras (Lucid, ZED), IMU/GPS (Applanix, SBG), RADAR (SmartMicro); NVIDIA Jetson AGX/NX, Neosys Nuvo, CANbus, UDP/TCP
- Languages:** Turkish (native), English (working proficiency)

Professional Work Experience

HAVELSAN – Robotics and Autonomous Systems

Ankara, Türkiye

ROBOTICS SOFTWARE ENGINEER

Apr 2021 – Present

- Owned end-to-end **C++ / ROS 2** perception, localization, sensor fusion, and autonomous navigation software across **three BARKAN UGV** platforms delivered for field operation.
- Developed and continuously validated autonomy software through **six months of field testing** on physical BARKAN platforms, debugging and iterating on the vehicle without a simulation environment.
- Designed, integrated, and deployed localization / state estimation: LiDAR-Inertial SLAM (LIO-SAM) with pose-graph optimization, plus EKF fusion of wheel odometry and GPS when available; **sub-20 cm** in GNSS-denied stretches. Streamed pose, velocity, and heading to the autopilot with or without GPS.
- Owned semantic perception for BARKAN UGVs (demo): DL segmentation, LiDAR-camera fusion, and semantic, voxel, elevation, and risk mapping for traversability-aware autonomy on unstructured terrain.
- Developed and provided costmap representations from semantic and geometric perception to the planning team, enabling traversability-aware path planning on unstructured terrain.
- Ran the full perception ML loop (DINOv2, SAM2, YOLO): domain data collection and training, mIoU/mAP eval, then **C++** deploy on Jetson AGX via ONNX Runtime / TensorRT and CUDA, tuned for latency and memory limits.
- Shipped a vision-only UGV follow-me system with real-time **SAM2** mask tracking at **25 FPS** (not a YOLO box tracker), then visual servoing to the segmented target.
- Developed UAV map-aided localization (video) with uncertainty-aware visual registration: **2.9–4.4 times** lower track error than constant-covariance baselines; **14–16 m** bias-free track MAE on **24–25 km** field sorties.
- Adapted and deployed Autoware on a Toyota Corolla (video), integrating perception, localization, and navigation for lane keeping, lane change, and obstacle avoidance.

TEKNOFEST Competitions

İstanbul, Türkiye

COORDINATOR

Jan 2019 – Apr 2021

- Coordinated competitions across 35 fields; trained and supported **1000+** participants, plus website, Q&A, and outreach that grew engagement and entries.

HARGEM

Kayseri, Türkiye

RESEARCHER STUDENT

Apr 2014 – Apr 2016

- R&D on high-speed wireless systems (modulation recognition, DSP/FPGA); **2 peer-reviewed papers** on OFDM and channel equalization (IEEE SIU, Journal of New Results in Science).

- Circuit design/test plus OpenCV detection, classification, and tracking prototypes.

Selected Research

- **GNSS-denied UAV localization.** Satellite-imagery visual localization with learned feature matching and registration; extended to uncertainty-aware vision-inertial geo-localization. SIU 2026; ICRA 2027 (under review).
- **AI perception on real robots.** DINOv2 / SAM2 perception pipelines, trained on domain data and deployed real-time on NVIDIA Jetson via ONNX Runtime and TensorRT (C++).

Education

Istanbul Technical University

M.Sc. IN ELECTRONICS ENGINEERING

Istanbul, Türkiye

Sep 2019 – Aug 2022

- Thesis: *Embedded AI Based Image Matching Method and Application in UAV*—onboard image matching for UAV geo-localization, later extended in fielded map-aided localization.

Nuh Naci Yazgan University

B.Sc. IN ELECTRICAL AND ELECTRONICS ENGINEERING (100% SCHOLARSHIP)

Kayseri, Türkiye

Sep 2012 – Jul 2017

- Thesis: *The Robot Can Follow The Smell*. GPA **3.24/4.00**, second in class; High Honor (x2), Honor (x2); TÜBİTAK 2209/A project.

Publications

Conference

- **Deep Visual Localization for UAVs in GNSS-Denied Environments Using Satellite Imagery**
E. Öksüz, T. Demirdal. 34th Signal Processing and Communications Applications Conference (SIU 2026).
- **Discrete Wavelet Transform Based Frequency Domain Channel Equalizer for OFDM Systems**
E. Öksüz, A. Özen. 24th IEEE Conference on Signal Processing and Communications, 2016.

Journal

- **A Comparative Performance Analysis of FFT Based OFDM and DWT Based OFDM Systems**
E. Öksüz, A. Altun, B. Ülgerli, G. Yücel and A. Özen. Journal of New Results in Science, No. 12, pp. 272–287, Sep. 2016.

Under review

- **Uncertainty-Aware Vision-Inertial Geo-Localization for GNSS-Denied UAVs**
E. Öksüz. IEEE International Conference on Robotics and Automation (ICRA 2027), under review.